

Aide à la réalisation / Présentation du projet

Données du projet				
Nom entreprise	Re : Voir SARL			
Nom du projet	BOLOTIE or « BOLO TIE »			
Résumé synthétique				
Durée (en mois)	18	À partir du	01/10/2026	
Coût du projet (€)	90 000	Aide demandée (€)	50 000	
Métier visé	Choisissez un élément. Autre :			
Chiffre d'affaires de l'entreprise	Répartition du chiffre d'affaires			
	% flux ¹	% stock ²	% interne ³	% externe
Année (n+1)				
Exercice en cours				
Année (n-1)				

¹ **Œuvres de flux** = journaux et magazines d'information, les jeux, les émissions de divertissement, la météo, le sport, le télé-achat, les messages publicitaires, l'autopromotion et les services de télétexte, les reportages internet, les jeux sérieux (serious gaming).

² **Œuvres de stock** = les œuvres cinématographiques ou audiovisuelles de type documentaire, fiction, œuvre d'animation, re-cr ation de spectacles vivants (existant ind pendamment de leur passage   la t l vision) ou magazine culturel r alis  majoritairement hors plateau, les webdocumentaires ou  uvres interactives   destination du second  cran, les jeux vid o de cr ation.

³ **Interne** = clients avec lien capitalistique direct ou indirect

Le dossier de présentation du projet est l'expression de la **qualité de la conception et de la maîtrise** du projet. Il constitue le support crucial pour permettre aux experts du comité de sélection de **comprendre rapidement et facilement** l'objet et les enjeux du projet.

Celui-ci ne pourra comprendre **plus de 20 pages**, hors annexes. Si des points d'ordre technique/organisationnel/économique/financier/environnemental nécessitent un développement particulier, ceux-ci doivent être traités dans une annexe spécifique. N'oubliez pas que **votre lecteur ne connaît pas a priori votre projet**. Soyez clairs et simples. Exposez les prémices, évitez le jargon et les anglicismes, définissez les acronymes. Illustrez votre document, légendez et commentez les illustrations. Rédigez une courte synthèse en ouverture de chaque partie. Aérez la mise en page en découpant le texte en paragraphes courts accompagnés d'un titre signifiant.

Le **dossier à constituer** doit être déposé en **version numérique avec un fichier par pièce et nommé tel que « NOMENTREPRISE – Intitulé.de.la.pièce »** (cf pièces complémentaires obligatoires) par mail.

Une attention particulière est portée aux projets en lien avec l'intelligence artificielle compte tenu des risques inhérents à cette technologie.

Les porteurs de projets concernés sont ainsi invités à communiquer des éléments précis qui concernent leur capacité à :

- Maîtriser la technologie et la donnée
- Evaluer les enjeux juridiques
- Evaluer l'impact social et environnemental
- Garantir la transparence de l'information auprès des utilisateurs et assurer la protection du public

En outre, il est attendu que le recours à l'IA soit pleinement justifié par rapport au besoin auquel il répond, comparativement à d'autres choix technologiques possibles.

Ces points d'attention viennent compléter les critères généraux d'appréciation de ce dispositif qui place la technique et la technologie au service de la création.

Les détails présentés ci-après sont donnés à titre indicatif et n'ont pas valeur d'exhaustivité.

MAITRISER LA TECHNOLOGIE ET LA DONNEE
- Données : origine, qualité, réduction des biais, méthodologie de préparation et constitution des datasets - Technologies : proportion de développements internes et externes, garanties d'autonomie et différenciation, possibilité de modification des éléments générés - Open source : recours et contribution à la communauté, type de licences appliquées - Sécurité : mesures de protection des données, cybersécurité et gestion des risques
EVALUER LES ENJEUX JURIDIQUES
- Propriété intellectuelle : méthodologie de respect des droits d'auteur sur les données et contenus en entrée et sortie de l'outil, rémunération des ayants droit - Données personnelles : conformité aux réglementations et garanties pour les utilisateurs
EVALUER LES IMPACTS SOCIAUX ET ENVIRONNEMENTAUX
- Emploi : impact positif ou négatif sur l'emploi, impact sur l'évolution des métiers, compétences et formation - Conditions de travail : enjeux de réduction de la pénibilité - Environnement : évaluation et réduction de l'empreinte écologique
TRANSPARENCE ET PROTECTION DES UTILISATEURS

- | |
|--|
| <ul style="list-style-type: none">- Confiance et fiabilité : efforts pour expliquer l'IA aux utilisateurs et garantir sa qualité- Protection des publics : mécanismes pour éviter les biais, risque de désinformation, protection des publics vulnérables |
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Partie A : L'entreprise, son actionnariat et son activité aujourd'hui (2-3 pages)

TO GET FROM RE :VOIR

<https://re-voir.com/en/pages/services>

A1 : Présenter les principales étapes de développement de l'entreprise, la répartition de son capital, le parcours des dirigeants, ses moyens humains (CDI, CDDU, etc.) et techniques (locaux, matériels, etc.), son/ses implantation(s) géographique(s), ses éventuels liens avec l'écosystème local (écoles, producteurs, etc.).

Two paragraphs, address those points below:

- Re:Voir founding, when, why, mission, its current state
- The services it provides and its unique position in the market for film, the type aid it has already received from CNC (in recognition of its unique place in the ecosystem of film in France or smeth)
- It's human power, a short bio of Pip, number of employees, Jim, Diana, tasks, stagiaires (list schools doing STAGE there such as the cinema department of Paris 8, etc) STATE that most of its special missions (such as Bolotie) is done through external collaborators that have become long-time contributors
- Its network of friends and collaborators, mention all the COOL places such as Jeu de Paume, Palais de Tokyo
- Any films schools universities?
- mention Film Gallery

Through the exhibitions hosted at the Film Gallery — inaugurated in 2005 in Paris as the first gallery in the world dedicated solely to experimental cinema — Re:Voir has promoted artists working at the intersection of film and other art forms, including visual artists such as Stéphane Marti and Jeff Scher, and artists connected to the Fluxus movement, such as Jonas Mekas. The Film Gallery also provides film projection equipment and maintenance services to major institutions, including the Centre Pompidou, the Jeu de Paume, the Palais de Tokyo, and MoMA, helping them screen works in their original film format rather than on video.

Re:Voir boutique has become recognized as the primary distributor and point-of-sale for Super 8 and Bolex-compatible 16mm film rolls in Paris. With nearly €80,000 of film sold per year, including nearly 600 Bolex rolls (which represent 4 out of 5 of the top-selling products), it is today the de facto destination not only for supplies, but for any kind of inquiry concerning small-gauge film cameras, for which Re:Voir also provides rental and repair services. Re:Voir has also observed this increased interest in analog filmmaking practices at a national level, with film-related activities now occupying a large portion of the company's overall revenue. The company is currently undertaking an expansion of operations to increase retail space for its film and camera business.

A2 : Décrire les activités actuelles de l'entreprise :

- ses principaux produits et services et/ou ses procédés propres,

get it from Re:voir
mention Film Gallery

- ses principaux clients,

get it from Re:voir

- le cas échéant des exemples de clients externes à l'entreprise,
- des exemples d'œuvres éligibles au soutien du CNC sur lesquelles la société a participé lors de l'année écoulée et si possible dans l'année à venir,
- la répartition de votre CA par typologie d'œuvres (long-métrage, série, documentaire, jeu vidéo, etc.) lors de l'année écoulée et si possible dans l'année à venir,
- le cas échéant les différentes activités exercées (postproduction, animation, édition, production, etc.) et proposer une ventilation des CA déjà réalisés et à venir.

get it from Re:voir, mention the aide it has received from CNC, mention Film Gallery

A3 : Présenter la stratégie environnementale de l'entreprise :

- existence d'une politique environnementale ou RSE (documentée ou en construction),

invent something (if this does not exist)

- identification d'axes prioritaires (énergie, mobilité et déplacements, équipements, numérique responsable, données, usages),

invent something (if this does not exist)

- modalités de pilotage et de suivi (référent, feuille de route, indicateurs, même simplifiés),

invent something (if this does not exist)

- pratiques liées aux équipements et infrastructures (durée de vie, mutualisation, réparabilité, fin de vie),

strong point, state Re:Voir missions of repairing, mention Christof?

- actions de sensibilisation ou de montée en compétences des équipes, le cas échéant.

mention again the wide circuit of external collaborators, give exemple, mention Film Gallery (ask Re:Voir)

Partie B : le projet et son environnement (12 pages maximum)

B1 : Le résultat du projet est-il : un nouveau produit ou service à destination commerciale ou qui permettra d'engendrer une augmentation chiffrable de votre productivité ? Une rupture dans ses méthodes de fabrication, par des méthodes organisationnelles ou des moyens de production qui n'étaient pas déjà utilisés dans l'entreprise ?

Description résumée en comparant avec la situation actuelle s'il s'agit de l'évolution d'un projet existant.

Throughout the 1980s and 1990s, the factories that once mass-produced affordable film cameras, such as the Swiss-based Bolex/Paillard in Sainte-Croix, were dismantled. Since then, producing a new film camera has proved a costly undertaking, and the remaining legacy camera manufacturers — Panavision, Arri, Aaton — eventually followed suit, releasing last-generation 16mm and 35mm professional cameras in the early 2000s before ceasing production entirely in the face of a forecast end to film production. However, as celluloid has proved its stubborn persistence in both the commercial and art film markets, the cost and availability of 35mm and 16mm cameras — already in the tens to hundreds of thousands of dollars for new models — continued to inflate, with demand emptying the used market as well. Today, newly manufactured boutique film cameras are extremely rare, their price sits in the tens to hundred-thousand-dollar range, and the existing supply of serviceable professional cameras continues to decline as the supply of spare parts dwindles. A few international initiatives have aimed to "revive film cameras" by developing new, sometimes less expensive models geared toward 16mm film production. These range from the Yolklab Y16 (yolk.org), the recently-announced Cinelux Sixteen (cineluxcinematools.com), and the Shtrihfilm K5 ([instagram.com/shtrihfilm](https://www.instagram.com/shtrihfilm)), which promise market prices in the 10,000–30,000-euro range. However, these remain, for now, small-scale prototypes, often impossible to buy, and prohibitively expensive for individual owners.

As a result, filmmakers wishing to use film for shooting, experimenting or educational purposes rely on older cameras, sometimes dating as early as the 1920s. And in the face of these challenges, one tool has remained steadily and increasingly called upon, and that is the Bolex H16 16mm camera, which was produced in massive quantities over multiple generations and which is widely considered to be one of the most reliable cameras ever produced.

The Bolex H16 Reflex was introduced in 1956, and underwent numerous iterations from the Rex 1 (1959) to the Rex 5 (1967) and bayonet-mount SBM (1970), before arriving at its final iteration, the electronically-powered and sound-sync Bolex EL in 1975. The spring-wound Rex / SBM models remain widespread and essential tools due both to their mechanical reliability and to their relative availability: it is estimated that over 100,000 spring-wound Bolex reflex cameras were produced. The electronic EL cameras, on the other hand, are increasingly in demand for their ability to shoot uninterrupted takes at sync speeds. However, these cameras were manufactured in smaller numbers and their internal electronics are highly prone to failure. (A [cinematography.com](https://www.cinematography.com) thread on the camera is titled, "Bolex EL Mk3 - The camera nobody wants to repair".) Boutique ventures such as OFG Customs (<https://www.ofgcustoms.com>) restore and modernize these models for contemporary production, but at a cost of well over \$5000.

Importantly, the use of these cameras remains remarkably widespread, used in both independent and arthouse cinema (Mark Jenkin's *Bait*, 2019; Marie Losier's *Peaches Goes Bananas*, 2024) and in major commercial productions (Bolex cameras appear in the camera

package on Tarantino's *Once Upon a Time... in Hollywood*, 2019, and on Josh Safdie's *Marty Supreme*, 2025). They also remain extremely common in film school inventories, where the Bolex serves as *the* standard model for learning to shoot film. Bolex cameras also make up the principle part of RE:VOIR's rental stock, and are the primary cameras requested by professional filmmakers and students for a wide range of creative projects (see fig. 1).



Figure 1 – Bolex models RX5 (left) and RX4 (right), available for rental at Re:Voir.

However, regardless of their widespread use of spring-wound Bolex Rex/SBM models, these models present significant technical restrictions that influence the type of images they can shoot and, ultimately, the range of productions that they can serve. Their main limitation concerns run time: the spring-powered motor must be rewound by hand and provides only around 28 seconds (at 24fps) of continuous running per wind. The available speeds are also limited to imprecise, non-sync values, selectable by a continuous dial that ranges between 8 and 64 fps. This evidently limits the length of an uninterrupted shot a filmmaker can achieve and, as a result, the type of sequences and movements these cameras can be used for. The unavailability of sync speeds further limits their usefulness for professional or narrative productions, even in a B-camera capacity.

BOLOTIE is a film camera accessory that seeks to significantly augment the capacities of Bolex cameras, and in doing so, expand the tool choice available to filmmakers. In simple terms, BOLOTIE replaces the internal motor mechanism with a digitally controlled external motor attached to the side of the camera (see fig. 2), turning any spring-wound Bolex H16 model into a fully motorized sync-speed camera. To date, no product exists on the French or international market that can provide the same functionalities. (See in B2 side-by-side comparison with the only two other products available, one of them discontinued today). BOLOTIE runs on an ESP32 S3 Nano microcontroller (2020), housed in a machined metal casing (rather than plastic) to ensure its reliability, longevity and suitability for professional projects. (The full list of components, materials, and the environmental considerations involved in their choice can be found in B6 and C2). As a new and technically innovative design, it has been conceived to satisfy three main criteria:

- **Creative:** It gives Bolex cameras capacities equivalent to advanced film cameras, with continuous sync speeds at any framerate up to 64fps. Additionally, it preserves and even augments the creative capacities that make the Bolex an essential

expressive tool: running in reverse, executing time-lapse sequences, and stop-motion animation. (The full list of its features can be found in B2).

- **Practical:** The resulting unit is compact, lightweight, portable, and can be easily attached to and removed from the camera.
- **Sustainable:** High-quality materials and lifetime software support, paired with open-source software, allowing the community to customize and add new functionalities according the needs of each user.

(add cool visuals)

Fig 2 – The BOLOTIE (left) and how it attaches on the camera (right)

BOLOTIE is thus a new product aiming to radically renew the ways Bolex cameras are used. It is meant to help create more visually complex films, permit experimentation with manual modes of film shooting, and become an indispensable tool for students, artists and film professionals alike. Faced with the impossibility of obtaining new film cameras, BOLOTIE improves the features and capacities of existing cameras, known for their longevity and durability, opening a new horizon of working with film.

B2 : Dans quelles mesures votre projet peut-il être qualifié d'innovant ou permet-il la création d'un avantage concurrentiel ? Exposer le marché cible (caractéristiques, typologie de productions, zone géographique...) et son appétence pour le projet (étude de marché, lettres d'intérêt, précontrats, etc.).

Décrire la concurrence existante et l'état de l'art technique, les forces propres de l'entreprise par rapport à cette concurrence. Autres projets connus (France, Europe, Monde) portant sur des sujets similaires.

evoque the interest we have “lettres d'intérêt” (include one email from one person who is very enthusiastic about BOLOTIE?)

Only two competing products in the global market

The Bolex is unique among film cameras in that it was designed from the outset to permit the addition of an external motor. Its exposed drive shaft and mounting points served to affix various external motors released by Bolex itself throughout the *(decade?)*, from the early MC-17 model to the late sync-capable ESM model. These motors, however, beyond being extremely difficult to find today, present numerous practical shortcomings: they are large, heavy, require a cumbersome external battery belt, most all of them are limited to single non-sync framerates, and none of them permit any features beyond the forward running of the camera. Except for rare cases, these motors are no longer in service.

However, the camera's compatibility with external motor drives allows to attach new, improved external motors. Filmmakers consulted while designing BOLOTIE pointed to two main accessories currently used with Bolex cameras: the Tobin crystal-sync motors and NCS's Revolution Motor (see side-by-side comparisons in Table 1 and Table 2 below). Neither offers the same combination of features as BOLOTIE at a comparable price, and BOLOTIE has also been designed to be considerably lighter and more flexible than either. As a result, BOLOTIE can be considered un produit innovant avec un avantage concurrentiel in the market of film cameras.

The first comes from Tobin Cinema Systems, a reputed Washington-based company that has manufactured professional film equipment since the 1970s. Their TTL Time Lapse Motor allows Bolex users to program time-lapse sequences and single-frame stop-motion animation. A separate motor, the TXM-20Ba, allows the camera to run at fixed sync speeds. Both products are now discontinued and resell for roughly €1,500 and €4,000 respectively on the secondhand market. Because each accessory covers only one mode, either time-lapse or standard-speed running, filmmakers must own both accessories and switch motors to change shooting function.

The second is the Revolution Motor (REV-Bx-100S), made by NCS Products in New York, a company specializing in designing gadgets for film cameras. The Revolution Motor allows both standard-speed running and intervalometer (time-lapse) operation from a single unit, and for many years was the main solution available for Bolex users. However, its current price – around €1,300 –, its weight – over 1 kg – and its size – separate motor unit and controller box – make it impractical for most types of shooting and a significant financial investment. At the same time, not being open-source or customizable, **its continuation is not certain, (as verified by personal communication with the company's main engineer when researching the possibility of developing BOLOTIE).**

BOLOTIE's advantages

BOLOTIE improves on the Revolution Motor in several ways. First, its speed range is wider, reaching up to 64 fps, compared to the Revolution Motor's 48 fps. BOLOTIE's speeds are also continuous, chosen freely by the user, rather than fixed in 1-fps steps.

Second, BOLOTIE provides a fully digital, user-friendly interface via an OLED screen, compared to Revolution Motor's thumbwheel rotary-switch interface that has no display and requires remembering a combination of parameters. Furthermore, BOLOTIE runs on an open-source software that allows users full control and customization of the motor's functions. BOLOTIE also includes a built-in light meter with an automatic exposure mode, a feature absent from both the Revolution Motor and the Tobin TTL.

Third, BOLOTIE can replace both Tobin accessories in terms of functionalities and it is designed to weigh under 0.5 kg, against the Revolution Motor's 1.1 kg. Its power system is also considerably simpler and modernized, accepting either an industry-standard LEMO 2-pin connector or a standard USB-C battery bank. Finally, even accounting for the high-quality durable materials chosen and the human labor involved in its production, BOLOTIE is expected to reach a market price under **€700 HT** per unit (see B3 for details on the economic model).

Table 1 - Side-by-side comparisons for the functionalities of the three accessories

	BOLEX sans accessoir	REV-Bx-100S (New York, USA)	Tobin TXM-20BA/E (Washington, USA)	BoloTie (Paris, France)
Drive shaft / camera compatibility	N/A	Toutes modèles (Rex 1 - Rex 5, SB/SBM) via 8:1 shaft	Uniquement Rex 4-5 et SB/SBM via 1:1 shaft	Toutes modèles (Rex 1 - Rex 5, SB/SBM) via 8:1 shaft
Run speed	12–64 ips, valeurs discrètes (réglage manuel, imprécis)	Oui, 1–48 ips, valeurs discrètes,	12, 15, 24/25, 30, 48/50 ips	Oui, 1–64 ips, valeurs continues (précis, l'utilisateur

		incréments de 1 ips (précis)		peut choisir toute vitesse souhaitée)
Crystal sync	Non	Oui, de 1-40 ips	Oui, à 12, 15, 24/25, 30, 48/50 ips	Oui, de 1-64 ips
Reverse	Rembobinage manuel au manivel	Oui, pleine vitesse ou image par image (conçu pour rembobiner pour double-exposition)	Non	Oui. Rembobinage automatique “last shot” pour double-exposition; Marche arrière “reverse filming” à 1–64 ips
Time-lapse / intervallomètre	Non	Oui : intervalle 1 sec – 9 heures. Exposition 1/8, 1/4, 3/8, 1/2 sec, ou exposition longue programmable jusqu'à 90 seconds. Enchaînement de préréglages disponible	Nécessite deuxième moteur, Tobin TCS TTL. Intervalle de 2 sec à 60 min. Deux expositions possibles : 3/8 sec fixe, ou exposition longue = 0.75 sec	Oui : intervalle sans limite, plusieurs expositions possibles (de 1/60 sec à 60 sec) ; Enchaînement de préréglages non disponible; éventuellement programmable via application
Stop-motion animation (image-par-imag e)	Oui : Exposition figée à 1/30 sec.	Oui : Exposition 1/8, 1/4, 3/8, 1/2 sec.	Oui : Exposition figée à 3/8 sec	Oui : Periode d'exposition variable de 1/60 sec à 1/60 sec
Light meter	externe	externe	externe	Embarqué — « mode auto » possible grâce au light meter intégré
Control	Contrôle mécanique (molette de vitesse + manivelle de remontage)	Contrôle numérique sans écran	Contrôle manuel (molette + boutons)	Contrôle numérique avec écran (boutons / rotary switch)
External control	Déclencheur à distance par câble mécanique	Boitier de contrôle peut être déporté par câble téléphonique RJ11	Precision speed control + run/stop via Tobin Milliframe Controller (DE9 connector)	Remote run/stop via 2.5mm GPIO connector Wifi+BLE intégré pour future contrôle par application smartphone

Table 2 - Side-by-side comparisons for the characteristics of the three accessories

	REV-Bx-100S (New York, USA)	Tobin TXM-20BA/E (Washington, USA)	BOLOTIE (Paris, France)
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Motor type	Moteur pas-à-pas (stepper motor), entraînement direct	Moteur DC 12,6V	Moteur pas-à-pas (stepper motor)
Poids	1,1 kg	N/A	~ 0,4 kg
Date	2005	2000s (production arrêtée depuis)	2027 (à venir)
Power connector	Prise Molex 2 broches ou XLR 4 broches	XLR 4 broches	LEMO 2 broches ou USB-C
Battery system			
Accessory anchor points	Non	Non	Oui
Price (in euros)	~ € 1280 (neuf)	~€ 3980–4760 (occasion/reconditionné)	€ 699 (neuf) [voir aussi B3]
Links	link	link , link	



Figure 3 – REV-Bx-100S (top left), Tobin TXM-20BA (top right) and BOLOTIE (bottom).

An additional important consideration for any modern camera is the potential for “building out” the rig, where modularity and a demand for multiple accessory anchor points has become standard among contemporary filmmakers. Especially for lightweight or handheld setups, it is common to mount battery, monitor, french flag, cinetape, accessory arms and FIZ accessories all on-board. This is especially critical for Bolex cameras, which are often chosen for commercial, fashion, sports and music work precisely for being compact and maneuverable. To this end, multiple companies already manufacture rigging accessories specifically for the Bolex (see <https://bolfix.ch/> and <https://www.ofgcustoms.com/bolex>). However, until now no one has been able to provide mounting points on the more convenient “operator side” without interfering with the camera crank and controls, and Bolfix’s side-plate risks excessive weight on the sensitive camera door mechanism (that also has to be removed for each roll change.)

Here, again, BOLOTIE manages what no one else has. By integrating all necessary motor and run/stop controls, the rest of the camera’s face can be used for accessory mounting points without interfering with camera function. The standard model BOLOTIE comes equipped with a single $\frac{1}{4}$ - 20 mounting point with Arri standard locking pins, which can be used to mount a battery system, for example. The larger accessory model, targeted toward professional users, populates the entire camera face with the Arri standard mounting hole pattern, with $\frac{3}{8}$ -16 and $\frac{1}{4}$ -20 holes and corresponding locking pins. With BOLOTIE, the Bolex can finally be rigged as a modern camera.

B3 : Quel est le modèle économique sous-tendant le projet et le temps nécessaire pour atteindre sa viabilité ?

BOLOTIE is intended to be a commercial product addressed to three main client targets:

- filmmakers and artists working with film
- rental companies
- film schools and institutions, such as Université Paris 8, La Fémis, L’Abominable, l’Etna

There are three phases in the development of BOLOTIE: prototype development, testing, and initial production. Each phase has its own costs (see Part C for details). The cost of components and labor involved in producing a single unit, when manufactured in batches of 50, is estimated at [€24,000, or €480/unit]. We estimate that selling **over 250 (check if 200 is better in the budget) units at** a market price of [€650] will cover the development costs and their life-time support in terms of software updates. Adjusting the market price is also possible to account for unforeseen circumstances, especially given that the closest competitor in the market costs approximately €1,300.

However, this model faces two main obstacles common to the development of new innovative products. First, achieving a low per-unit cost requires manufacturing in minimum batches of 50 to 100 units, which demands a substantial upfront investment before the first sales are even made. Second, reaching the target of **250** units requires a sustained and effective promotional campaign, the cost of which precedes and does not depend on unit sales. Together, these two factors create a funding gap between the investment required to launch production at viable scale and the revenue that only follows once that production

exists. Support from the CNC is therefore indispensable to allow the project to reach the scale at which it can become self-sustaining. Beyond 250 units sold, BOLOTIE requires no further external funding to continue production and maintenance of the produced units.

Sales beyond 250 units will represent net profit, which will support the design of new, improved accessories for film cameras, drawing on the technical expertise gained from designing BOLOTIE and the feedback received from its users. Finally, additional income will be generated through a bundled "Bolex + BOLOTIE" rental offer for artists who already rent Bolex cameras through Re:Voir ([how many for last year's activity?]), though this will represent a very small percentage of overall income, mainly covering administrative expenses (see Part C for details).

B4 : Décrire la stratégie d'accès au marché et les moyens mis en œuvre pour sa promotion.

BOLOTIE will be promoted through two platforms. First, it will be advertised to Re:Voir's existing client circle, through its website and newsletter, as a new product available for purchase or for rent alongside Bolex rentals. Re:Voir being one of the few shops in Paris that sells film stock for shooting, this advertising campaign is expected to reach a significant share of the community engaged in film shooting in France, as Re:Voir already has a wide network of artists visiting the shop on a daily basis. Informal communication with them about BOLOTIE has convinced us that the product is much desired and will be well received. Second, BOLOTIE will be advertised through a dedicated website (add link), designed specifically for this purpose, intended to reach an international clientele, as BOLOTIE aims to become the international gold standard for Bolex shooting.

Further, BOLOTIE will be advertised in three phases, following its production planning. To design the promotional campaign in France and internationally and sustain a regular presence on social media a communication coordinator will be hired (see part C). During the development of the prototype, a clear and captivating visual identity will be designed, using a dedicated logo as well as photographs of the unit, its design and its schematics (see a first version of the promotion visuals here : add link). After the testing phase, videos shot with BOLOTIE and the testimonies of the testers will also be published, to provide live feedback from real test cases. A demonstration event will also be scheduled at Re:voir to present BOLOTIE and let people handle and familiarize with the unit.

Once testing proves successful, a well-known director of photography will be hired for a promotional video, to increase visibility in the wider professional film community at an international level. In parallel advertisements will be published in relevant publications such as l'Association Française des directrices et directeurs de la photographie Cinématographique (AFC), American Cinematographer, cinematography.com, and NoFilmSchool (nofilmschool.com) (see the full list in C2).

Finally, once in production, specific actions will be taken, such as participating in trade fairs, exhibitions, and festivals dedicated to experimental film, for example the Micro Salon AFC in Paris, the Cine Gear Expo in Los Angeles and the Punto de Vista in Spain to present BOLOTIE in person to professionals and the artistic community, both in France and internationally. In this, the open-source element will be a key feature to use in the promotion to build interest and further engage the artistic communities that wish to personalize the equipment.

Points to consider for adding here or elsewhere

Add in info about expanded Femis programs (like with L'Abominable) and film school investment, like Wesleyan.

- A pilot loan program with film schools, promo with film schools
- Cite specific artists we will reach for testing (like Guillaume or people we know through revoir)
- Film Labs network. Planned promotional campaign featuring known cinematographers and filmmakers. Advertising campaign in relevant trade publications (AFC, American Cinematographer, cinematography.com). And outreach campaign to get articles in related publications (NoFilmSchool, Wired, etc.)

B5 : Quelle est la place du projet dans la stratégie de développement de l'entreprise ? Quelle est la dépendance de l'entreprise au succès du projet ?

Currently, Re:Voir is one of **three shops** in France that sell film rolls not only for **Bolex but also for Super 8, Double 8, Single 8, Double Super 8, and 35mm cameras (Beaulieu, Nizo, and Canon, among others)**. It also rents 8mm and 16mm cameras, lenses, and accessories. Over the years, it has built a stable community of filmmakers among its clients and collaborators, many of whom have become long-term collaborators.

Through the exhibitions hosted at the Film Gallery — inaugurated in 2005 in Paris as the first gallery in the world dedicated solely to experimental cinema — Re:Voir has promoted artists working at the intersection of film and other art forms, including visual artists such as Stéphane Marti and Jeff Scher, and artists connected to the Fluxus movement, such as Jonas Mekas. The Film Gallery also provides film projection equipment and maintenance services to major institutions, including the Centre Pompidou, the Jeu de Paume, the Palais de Tokyo, and MoMA, helping them screen works in their original film format rather than on video.

The idea and the very possibility of making BOLOTIE are a direct result of **decades** of Re:Voir working closely with artists and technicians, and gradually gathering collaborators with the technical expertise, such as repair technicians, schematic designers, electronics and software engineers, who are required for the development of such complex product. Throughout the years, we have observed that several artists working with Bolex cameras are dissatisfied with the technical limitations of the cameras. At the same time, the products available do not seem to be attractive, as they are either difficult to obtain or too expensive and inflexible. (Add more stories about repair and the “broken motor” that bolotie would fix?) As a result, most of them would be willing to rent or buy a new accessory that would help overcome the cameras' technical limitations. BOLOTIE's design details stem from this first-hand feedback coming directly from filmmakers. As such, we consider that no comparable initiative is currently in progress, either in France or elsewhere.

Concerning the over strategy of the enterprise. BOLOTIE will be integrated into Re:Voir's current rental offer and will also be sold separately. This will achieve three objectives:

- **First**, it will provide a much-needed solution for existing clients working with film.

- **Second**, it will attract new clients seeking to rent cameras, as the bundled Bolex + BOLOTIE will provide unique features and shooting possibilities compared with a standard Bolex rental from other sources.
- **Third**, as BOLOTIE becomes known and is progressively adopted by rental companies and film schools, hopefully at an international level, Re:Voir will benefit from the prestige and reputation of being the company that launch it, which will further strengthen its overall commercial activity. Further, an extensive testing phase for each unit will ensure the quality of the resulting equipment.

B6 : Comment les enjeux environnementaux ont-ils été intégrés dans la conception et la réalisation du projet afin d'en limiter les impacts ?

Environmental considerations have been present throughout BOLOTIE's design process, in the choices of components and production locations as well as in the overall conception of the project. Our approach is detailed in two axes:

Sobriety, logistics and production locations

As explained above, BOLOTIE seeks to extend the working life of existing film cameras, such as the Bolex H16 and Rex family, equipment manufactured decades ago. This is not just a practical choice due to the absence of new cameras, but adding functionalities to what already exists, rather than manufacturing new equipment, is one of the most direct forms of sobriety face aux enjeux environnementaux. This also fits into Re:Voir's broader mission of bringing old film equipment back into service. A main reason for the dysfunction of Bolex cameras is the failure of their spring-motor, one of their most fragile parts. BOLOTIE replaces the internal motor mechanism with a new external one, meaning it can restore functionality to Bolex cameras that are currently out of use for this reason. As a result, BOLOTIE not only improves the performance of existing equipment, but also restores old cameras to a working state, saving them from being discarded or dismantled for parts.

The main negative impact concerns shipping. As BOLOTIE involves a custom electronic design of an ESP32 chip attached to hundreds of pieces, cables and connectors (a **printed circuit board – PCB**), the main manufacturer chosen is based in China (JLCPCB). Their all-in-one solution enables us to manufacture something quite complex at a low enough cost. European alternatives were examined and rejected on cost grounds as the per-unit cost of the PCB (**was 5 times higher with a European manufacturer**) (**rose from [€...] in China to [€...] with a European manufacturer**), a difference that would have made BOLOTIE's target price of under €700 unachievable.

Producing in large batches of over 50 units or 100 units becomes thus all the more crucial, as it will not only allow us to reduce the market price, but also concentrate shipping into fewer, larger, better-optimized orders rather than frequent small shipments, directly reducing the carbon cost of transport per unit. Reaching this batch scale requires upfront capital that unit sales alone cannot initially provide (see Part C). CNC's support in closing this funding gap therefore contributes directly not only to the project's viability but also to its environmental performance.

Where the cost difference is less prohibitive, local sourcing is prioritized even if prices are higher. The camera-mounting case, for example, will be produced by [local manufacturer

name], a France-based supplier, chosen specifically because the case is among the bulkiest and heaviest components of the final product, meaning its transport footprint matters disproportionately. Splitting production this way, global manufacturing for compact, dense electronics and small machined parts, local sourcing for bulkier mechanical components, is meant to optimize the total shipping footprint. **Finally, in case BOLOTIE reaches sales allowing it to maintain its market price even with higher production costs, changing the CBC production from China to Europe will be examined.**

PCB, CNC, plastic covers, motor

Durability and longevity

However, we consider the most significant environmental contribution to be durability. Every major design decision in BOLOTIE has been made with the full life cycle of the product in mind. First, the body is made of anodized aluminum rather than plastic. Even if metal costs more and is heavier to ship, its resistance to wear and damage extends the unit's usable life far beyond what a plastic case would allow. The other custom components (shaft, buttons) are similarly machined from the most durable material suited to each part, hardened stainless steel, POM, or anodized aluminum, chosen specifically to reduce waste from parts wearing out and needing replacement. Additionally, the durability of these pieces is crucial to also avoid re-shipping of materials, and thus limiting the carbon footprint.

Second, even if that increases the overall production cost, the internal electronic components used are all mid- or high-tier, chosen over cheaper alternatives that may prove less reliable over time. All parts sit on a single circuit board, built from RoHS-certified components — cleaner and safer to produce and dispose of, with automotive-grade connectors throughout, which means higher production yield and a longer guaranteed lifespan. Additionally, the entire product can be taken apart and serviced. Also, rather than being permanently sealed, BOLOTIE uses gaskets and fasteners that can be opened to repair or replace individual components, rather than requiring the whole unit to be discarded if one part fails.

Third, software support is meant to guarantee proper functionality for several years. Bugs will be addressed through regular updates, and the software being open-source means that the product does not depend on the survival of any single company or team to remain usable and maintainable, directly addressing the sustainability risk earlier identified regarding the two main competitors. Together, these choices are intended to make BOLOTIE a once-in-a-lifetime purchase, usable for decades, rather than a product with a planned or de facto short replacement cycle.

Éléments attendus :

- choix de conception visant à réduire les impacts dès l'amont (sobriété, optimisation des usages, organisation des déplacements et de la logistique, limitation des flux et du stockage),
- justification des critères de sélection des matériaux et services mobilisés,
- prise en compte du cycle de vie des équipements et services mobilisés,
- comparaison, lorsque pertinente, de scénarios techniques alternatifs et justification des choix retenus,
- identification des impacts évités ou réduits et inscription dans une logique d'amélioration continue.

Partie C : l'exécution du projet (8-9 pages maximum)

C1 : Compétences et qualification de l'entreprise pour la conduite du projet. En cas de recrutement quelles seront les missions confiées à vos nouveaux collaborateurs ? Fournir les fiches de poste correspondantes et les CV des principaux participants en annexe.

The recent years have seen a sharp increase in interest in film, for both photography and cinema, at a global level. Kodak, the main supplier of film rolls, has stated that last year they sold more motion-picture film than at any point since 2014, and that they need to restructure their production chain to respond to the increased demand.

Re:Voir is a world-recognized expert in the small-gauge analog camera market, and since the closing of the Kodak Comptoir on rue Poncelet in 2010, the centrally-located Re:Voir boutique has become recognized as the primary distributor and point-of-sale for Super 8 and Bolex-compatible 16mm film rolls in Paris. With nearly €80,000 of film sold per year, including nearly 600 Bolex rolls (which represent 4 out of 5 of the top-selling products), it is today the de facto destination not only for supplies, but for any kind of inquiry concerning small-gauge film cameras, for which Re:Voir also provides rental and repair services. Re:Voir has also observed this increased interest in analog filmmaking practices at a national level, with film-related activities now occupying a large portion of the company's overall revenue. The company is currently undertaking an expansion of operations to increase retail space for its film and camera business.

BOLOTIE therefore comes at a key moment for cinema shot on film, since, as stated above, it will not only improve existing cameras, but it also has the ability to restore functionality to cameras currently out of use due to spring failure — the main cause of breakdown in old Bolex cameras — thereby increasing the overall pool of functional film cameras. We thus expect its development to improve shooting practices, and to encourage artists and students who are currently held back by technical difficulties or the high entry cost to start working with 16mm. To give an example, a "broken" Bolex that could become operational again with BOLOTIE can currently be acquired for less than €500.

With over 30 years of experience working with film material, Re:Voir has gradually built a wide international network of client artists and collaborators, and become a reference point for the film community (an overview of Re:Voir's network can be found here: <https://re-voir.com/en/pages/liens>). It has also gathered the technical expertise needed for developing, testing, and efficiently promoting new film camera equipment. BOLOTIE will be developed in collaboration with two engineers who are long-time collaborators of Re:Voir: Nathaniel Draper and Antonios Lagarias.

Nathaniel Draper (add your stuff) (full CV attached). Nathaniel apprenticing with Christophe for Bolex repair

Antonios Lagarias will be responsible for software design and maintenance. With a degree in engineering from Aristotle University (Thessalonique) and a master's degree in film from the ENS-Ulm (Paris), he has been developing software for research purposes during his PhD in digital humanities, in collaboration with Rennes 2 University and the Université de Montréal. Lagarias has been working on an initial version of the code required to run BOLOTIE, consisting of over 2,000 lines of code written in C++. Starting as an intern in 2018, he has been collaborating with Re:Voir ever since, developing software for exhibitions hosted at the Film Gallery. (full CV attached)

C2 : Description des ressources et moyens techniques mobilisés. Justifier les choix techniques et organisationnels. Joindre tout visuel permettant une meilleure compréhension de votre projet (schéma, synoptique, plan côté, etc.)

Transmettre le calendrier détaillé du projet avec, le cas échéant, un lotissement en sous projet sous la forme d'un rétroplanning.

Calendrier détaillé du projet

Initial tests and experiments on BOLOTIE started **last year**, carried out between Re:Voir and the two developing engineers to examine and prove the project's feasibility (phase 0). The results of these experiments are very promising. After several sessions of trial and error, updates in the PCB schematics, the components and the overall design of the unit, we have confirmed that BOLOTIE can reach crystal-sync speeds at all of the modes envisioned. We thus decided to proceed with its commercial development. BOLOTIE's development can be divided into three broad phases: prototype, testing, and production of the initial batch.

Below is the estimated calendar:

December 2025 - Septembre 2026: Phase 0 – Examination de la faisabilité du projet

Over the last year, Re:Voir, in collaboration with the two developing engineers, Nathaniel Draper and Antonios Lagarias, conducted a series of experiments in electronics and software design to examine the feasibility of BOLOTIE. Different combinations of electronic chips, stepper motors, drivers, and software were tested to verify whether the target speeds could be reached at the desired accuracy. In parallel, a first version of the case was designed to verify that it could be accurately mounted to the Bolex H16 and Rex models. At this stage, the case was 3D printed in plastic, since several experimentations were required to verify the exact dimensions, needing a precision to the exact millimeter

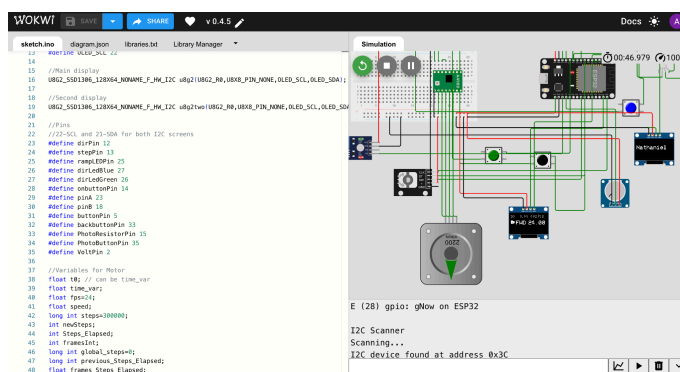
(**add some technical difficulties for the case here**) (see case design fig below, fig 5 and on the website)

First, we designed a virtual model, to mainly verify the software and the various connections required (see fig. 4). A physical version was then built on a breadboard to test different combinations of parts and power supplies. Once the combination of components was finalized, we designed a first version of the PCB (see fig 4.) and ordered a batch of five units from JLCPCB. This first design presented several issues in the electronic connections between the pieces. After several iterations of trial and error, we designed a new version for the PCB and ordered a second batch of five units with the updated components (see the final components list in Table 3). The new PCB succeeded in running a Bolex camera at the highest desired speed (64 fps, see video on the website). This success serves as a proof of concept for BOLOTIE, allowing us to move to the next phase : fine-tune the software, the board connections, and the case design to develop a fully functional prototype and apply for financial support.

At the same time, these initial technical experiments were coupled with a first unofficial promotion, discussion with Re:Voor's existing clients to examine whether this project responds to the needs of the filmmaking community. Initial reactions were enthusiastic, with quickly word spreading on its own. Several filmmakers reached out to us to ask for updates, to make sure they can order a unit as soon as possible, or to act as test user for the testing phase (see a list of emails and messages in the accompanying documents as proof of interest).

Finally, since the project started with open experimentation, the cost associated with phase 0 mostly concern the 10 PCB units, the testing of different components and buying the equipment required to print several of plastic cases to verify the design is an exact fit on the cameras, before upgrading to the more costly metal manufacturing.

Fig4. Phase 0 : The virtual model (left) and the breadboard physical version (right)



add one pictures of breadboard production

Table 3 - The full list of components for Bolotie after phase 0

add here

Fig 5 - the case design?

Octobre 2026 - Décembre 2026 : Phase 1 – Développement du prototype

After the project's feasibility has been confirmed in phase 0, we proceed by assuring the full functionality of the prototype. This involves several iterations of testing and feedback between Re:Voor and the developing engineers to: a) optimize the design of the digital menu and the navigation for users; b) test different cables and connectors to verify the stability of the power source and the overall performance of the unit; and c) verify the correct functionality of all modes (run forward, run in reverse, time-lapse, stop-motion, and auto mode with light-meter calculation). Testing units will be ordered in batches of 5 (2 or 3 orders are estimated) and the PCB schematics will be updated until no more issues can be detected (see fig6). This phase is estimated to last approximately three months, and its main costs involve the cost of components and new PCB manufacturing, specialized tools, and the engineers' fees.

Fig 6 the PCB schematics

During this phase, a first official promotion campaign will also be launched through the project's dedicated website and social media. An important part of the online promotion will be entrusted to an external collaborator, who will be charged with designing and sustaining the promotion strategy that will take place in Phase 2, both in France and internationally.

Finally, this phase also includes some administrative costs, including accounting, ordering, and website updates. Overall, BOLOTIE is expected to increase the workload of Re:Voir's existing employees on average by 10–15%. This does not justify the creation of new positions. However, to account for this increased workload, part of Re:Voir's head of sales, Jim Stickel's, time will be reallocated to the project. BOLOTIE funding will cover a percentage of Jim's salary and charges accordingly (see budget). Each phase represents different workloads, but for the uniformity of the budget a standard funding of 10% for the total duration of the project (18 months) was preferred over dynamically adjusting this percentage each month. Similarly, 10% of the accounting costs attributable to [partner's name], Re:Voir's accounting partner, will be covered by BOLOTIE funding, following the same logic (*clé de répartition (cost-allocation key)?*).

Janvier 2027 - Avril 2027 : Phase 2 – Testing auprès d'artistes concernés

At the end of Phase 1, the design of the PCB and the case will be finalized. Five units will then be produced and distributed to collaborating artists who have already expressed interest in testing BOLOTIE under real shooting conditions. This phase will last approximately four months. It will consist of several shootings followed by feedback discussions between the developing engineers and the collaborating artists, in order to identify design errors and software bugs. Careful attention will be paid to anticipating all possible use cases, to ensure that the final design remains functional even under difficult shooting conditions (continuous running, battery issues, etc.). After testing, BOLOTIE v1 version will be considered final and the software will be published on a dedicated github page.

This phase will also generate promotional material. Images of BOLOTIE “in action”, as well as videos and images shot with the Bolex + BOLOTIE rental offer, will be gathered from the collaborating artists, providing a first pool of testimonies to be uploaded to the website and used in social media and advertising. Finally, the artists' feedback concerning further functionalities will be documented and eventually published as github issues to further involve the community and lay the way for the development of a v2 software version in the future, by either Re:Voir or by artists wishing to customize and personalize the open-source software.

In terms of cost, the artists will test BOLOTIE on a voluntary basis, which will significantly reduce the overall fees. Their availability also demonstrates the community's enthusiasm for the new equipment. The main costs will involve the engineers' fees and the cost of components for the testing units. Phase 2 will also include a wide promotional campaign on social media, announcing the arrival of BOLOTIE and gradually publishing the promotional material gathered so far. This campaign will be led by the external collaborator engaged in the project since the end of Phase 1, who will manage BOLOTIE's social media presence, coordinate the rollout of testimonials and footage from the testing artists, and pursue coverage in specialized outlets, including forums and publications read by the film community, such as cinematography.com, and trade press such as American

Cinematographer or British Cinematographer, alongside analog-photography community sites (EMULSIVE, Kosmo Foto, 35mmc) (see full list below). This campaign is also expected to increase the administrative workload consisting mainly of responding to inquiries about the upcoming product and registering pre-orders. The rest of the administrative and accounting costs are expected to remain similar to Phase 1, so, as stated above, the uniform funding of 10% for all phases will not be modified.

May 2027 - September 2027 : Phase 3 – Production and promotion of first batch of units (50-100)

After BOLOTIE v1 has been extensively tested and all identified issues corrected, we will proceed with producing the first commercialized batch. Depending on the funding secured and the level of interest and pre-orders raised during Phase 2, this batch will comprise between 50 and 100 units. This first commercialized phase of the project will also serve to evaluate and optimize all aspects of the production and client support chain.

Over five months, units will be assembled and individually tested by the developing engineers before delivery to clients. A public live demonstration will be hosted at Re:Voir to present the product and provide hands-on testing. It will also showcase the potential of a Bolex + BOLOTIE rental bundle for professional and educational shooting contexts.

To further promote BOLOTIE, a promotional video featuring a professional filmmaker will be produced and shared on social media and the product's webpage. The advertising campaign will be extended with further social media announcements, and in-person presentations at (see full list below) :

- Trade fairs to reach professional cinematographers and industry buyers.
- Festivals to reach BOLOTIE's core artist audience.
- Film schools to present the equipment to students

The main objective of Phase 3 is to examine, reassess, and optimize all logistics involved in production, assembly, and shipping, adjusting the process wherever improvements can be made, before the project can be considered complete. The board alone comprises nearly 100 separate components. There are also 8 discrete CNC metal parts in each unit, and over ten external components that need to be integrated. To produce BOLOTIE, we'll collaborate with four custom suppliers (PCB, CNC, plastic covers, motor), plus all the discrete off-the-shelf components — requiring significant coordination and follow-up. During this phase, we anticipate leveraging the detailed knowledge the engineers have developed throughout the design process by shifting their role from design to production management. Costs for Phase 3 therefore include the engineers' fees and manufacturing costs for the units. Promotional expenses are expected to cover the collaborator's fees, as well as travel costs and hosting demonstrations at each festival, school, and fair.

Phase 3 will therefore end with a comprehensive evaluation of the project that will determine the details for its continuation: the size of new production batches, the planning of new orders for parts and materials, adjustments needed for the promotional strategy in France, new strategies for international clients, the planning of software updates, the communication channels needed to help users customize the equipment, and the ongoing support to be established for technical and software issues.

October 2027 – March 2028: Normal commercialization (?)

By this point, we can consider that the product has reached its maturity. The production and support chain is well established, and the ordering and shipping logistics are optimized. We can then move to the final phase: the normal commercialization of the product, in the hope of reaching over 250 sales and achieving a return on investment.

This target of 250 units is realistic yet ambitious. Even though over 100,000 Bolex cameras were originally produced and a large number remain in circulation, many of them present technical issues beyond the spring-motor failures BOLOTIE can fix. Furthermore, even though Bolex cameras are used in commercial cinema, the community of artists who own and operate 16mm Bolex cameras remains fairly small, consisting mainly of art et essai and experimental filmmakers and students. For many of them, even at its low market price, BOLOTIE may represent a significant investment that many are not immediately willing to make.

However, even if this target is not immediately achieved, we believe that BOLOTIE's contribution to the filmmaking community outweighs any profit concerns. BOLOTIE will enable broken Bolex cameras to be used again, while offering richer and more complex functionality for shooting on 16mm. As a result, we hope BOLOTIE will help artists create more visually rich films shot on film, while also lowering the entry cost to 16mm film production and thereby providing essential revenue to the film market as a whole. Further, if the target is reached and profits are made, they will be directed toward developing a BOLOTIE v2, using feedback gathered during the testing period and initial commercialization. Profits will also fund the development of further accessories, as the process of designing BOLOTIE has shown that there are several aspects of film cameras that new technologies and software can meaningfully improve, not only in Bolex cameras, but in other old film cameras as well, such as Beaulieu, Nizo, or Eclair models. The technical expertise gained through BOLOTIE and the collaborations established is thus expected to be channeled into an open and ongoing research on repairing, improving, and promoting film cameras.

Finally, BOLOTIE aims to demonstrate how new technologies can radically extend the life of old equipment, and to respond to the renewed interest in film shooting observed in recent years. As an example, the software running BOLOTIE uses in its current version only 20% of the ESP32 chip's capacity, leaving considerable room to add further personalized modes tailored to each user's needs. **BOLOTIE is, at its core, a small object carrying a large ambition: to prove that the future of film may not be built from new cameras, but from the ingenuity, care, and community built around the ones that already exist. (end on some bullshit)**

Final visuals?

En cas de :

- Création de poste, fournir un organigramme exhaustif des salariés permanents de votre société avant/après embauche,

No new fiche de posts is prévu for the development of BOLOTIE. The collaborating engineers will be paid sur devis (see C3). To account for the increased administrative costs, an 10% of Re:Voir's current head of sales, Jim Stickel, will be allocated to the needs of the project. The reallocation is stated with a **lettre de** , (see supporting documents) and represents **...euros for 18 months (for a total of ...)**

- Formation, préciser les nouvelles compétences acquises par une partie de votre personnel qui permettront la réalisation du projet et son exploitation, en cas de formation interne transmettre le plan de formation.

Nothing to mention here.

Antonio's coding classes
Nathaniel apprenticing with Christophe for Bolex repair
Both of us working with Palais de Tokyo geeks on AV inventions
Collaborateurs de longue date

- Changement notable d'organisation ou de procédés, dans quelle mesure votre démarche engendre une rupture dans vos méthodes de fabrication ? Quelles sont les difficultés techniques et/ou méthodologiques présentes dans le cadre du projet ? De quelle(s) façon(s) allez-vous les résoudre ? fournir un schéma "avant / après" faisant figurer de façon non ambiguë l'investissement faisant l'objet de la demande d'aide,

Nothing to mention here.

- Nouveau produit ou service, **décrire les éventuels verrous technologiques identifiés et justifier les choix technologiques effectués**. Si les dépenses incluent :
 - Un partenariat avec une autre PME et qu'aucune des sociétés ne supporte à elle seule plus de 70% des coûts admissibles : indiquez les motivations à l'origine de ce rapprochement, les compétences de l'entreprise partenaire et la répartition du projet entre les deux structures.

No partnership is deemed necessary at this stage. With over three decades of experience, Re:Voir is uniquely positioned among companies active in film equipment and cameras to lead and sustain the development of camera accessories in France. Re:Voir will draw on its network of collaborating structures, such as L'Abominable, the Centre Pompidou, the Jeu de Paume, the Palais de Tokyo, and MoMA, (see the full list here [link]), to promote BOLOTIE and co-host hands-on demonstrations during Phase 3.

Un reversement public des résultats du projet : indiquez les modalités de diffusion (logiciel libre, type de licence appliquée, animation de communauté, conférences, etc.),

As stated above, the software required to run the equipment will be published on the project's GitHub page under the MIT license, which freely allows copying and modification. This means that every user can potentially customize their equipment, either on their own or with the help of community support. The hardware is also designed to facilitate community modification and development, with 2 channels of pre-configured GPIO exposed via the front-facing 2.5mm jack (allowing variable input and output controls), and integrated WiFi and Bluetooth capabilities that will be further developed into an open-source control app. This

aligns with Re:Voir's vision of supporting and spreading the use of film cameras while further involving the artistic community in developing tools for its own needs. The hardware is designed to be easily detached and examined if needed, though the original hardware design schematics will remain the property of the developing team.

- Participation à des foires et salons, décrire le(s) salon(s) visé(s), les personnes mobilisées au sein de votre entreprise et les moyens mis en œuvre pour la préparation, la tenue et le suivi de votre participation,

The promotional strategy will be assigned to an external collaborator, who will design and execute the advertising campaign both in France and internationally. They will need in-depth knowledge of the analog camera and film network, and be able to identify and reach out to structures that could be interested in BOLOTIE — artists, rental companies, film labs, and archives. The collaborator will be paid on an invoice basis **at a rate of €3,500 per phase (€10,500 total)**. When in-person demonstrations require technical expertise, these will be carried out by the collaborating engineers and Jim Stickel, as head of sales, depending on availability. The overall efficiency of the advertising campaign will be evaluated at the end of each phase by the developing team and Re:Voir, with adjustments made where needed.

The trade fairs targeted are Micro Salon AFC (Paris, held annually by the Association Française des Directeurs de la Photographie Cinématographique) and Cine Gear Expo (Los Angeles), both major venues for reaching working cinematographers and industry buyers. **Concerning festivals, a French festival, Ann Arbor Film Festival and Punto de Vista (experimental/analog-focused festivals)**. At the end of each action, a post-event report will be asked to update a contact database that is gradually built from people and structures interested in BOLOTIE.

Finally, **Re:Voir is uniquely positioned to generate widespread awareness from the outset of the project, thanks to its activities as a film publisher and distributor active in the field for over 30 years. Re:Voir regularly communicates press releases to an active subscriber base of over 10,000 film professionals, institutions and enthusiasts worldwide.**

The promotional actions are detailed below, by phase:

- **Phase 1:** In-person communication with existing Re:Voir clients during their visits to the shop. Announcement on the website and social media about the upcoming project. Research to identify people and structures who could be interested.
- **Phase 2:** Newsletters, emails, and announcements about the product, including its full technical details. Gathering material (images and photos) generated by the testing artists. Preparing the advertising campaign (writing ads, purchasing ad space, designing visuals). Coordinating the production of a promotional video.
- **Phase 3:** A second wave of newsletters, emails, and announcements about the product. Reaching out to identified artists and structures. Sending targeted ads to magazines. Press kit. Finalizing the promotional video and identifying the best platforms for its distribution (web, TV) in France and internationally.

- Dépenses de promotion, détaillez les actions envisagées et la motivation de l'entreprise pour les mettre en œuvre.

Promotional actions, both online and on-site, are as follows:

- **Developing a dedicated product website for Bolo Tie, with the aid of a web developer**
- Publications on Re:Voir's website, blog, and in-store promotion
- Publications on BOLOTIE's dedicated webpage
- Social media publications
- In-person demonstrations at Re:Voir and its network of partner structures, such as L'Abominable, the Centre Pompidou, the Jeu de Paume and the Palais de Tokyo
- In-person demonstrations and promotion at film schools and universities, including La Fémis and Université Paris 8, **l'Etna**
- A professional promotional video featuring [name]
- Targeted advertising in:
 - **French outlets:** La Lettre de l'AFC (afcinema.com),
 - **International outlets:** American Cinematographer (ascmag.com), cinematography.com, NoFilmSchool (nofilmschool.com), Wired (www.wired.com), EMULSIVE (emulsive.org), Kosmo Foto (kosmofoto.com), 35mmc (www.35mmc.com/)
- Presence at festivals and trade fairs:
 - **French:** Micro Salon AFC (Paris)
 - **International:** Cine Gear Expo (Los Angeles); Ann Arbor Film Festival and Punto de Vista (experimental/analog-focused festivals).

The overall cost of these actions is estimated at **€10,000** excluding the fees for the promotional collaborator detailed above.

C3 : Présenter les compétences des partenaires, prestataires ou fournisseurs essentiels du projet, les modalités de collaboration et les ressources mobilisées de leur côté, et notamment la manière dont ils s'inscrivent dans votre démarche environnementale.

BOLOTIE's design will be entrusted to two collaborating engineers, who will assume the following tasks:

- **Phase 1:** Software, board, and case finalization; testing of connectors and power stability; verification of all operating modes.
- **Phase 2:** Distribution of testing units to collaborating artists; feedback collection; correction of design and software issues under real shooting conditions.
- **Phase 3:** Assembly and individual testing of the first production batch; oversight of manufacturing quality and shipping logistics.

They will be paid on the basis of an invoice, **€3,500 per phase (for a total of €10,500)**. Their technical expertise has already been demonstrated during the initial experimentation carried out to establish the project's feasibility (Phase 0), and their CVs are attached in the accompanying documents.

As stated above, there are two main production locations:

The PCB will be manufactured by the China-based JLCPCB, one of the world's largest PCB and electronics manufacturers, founded in 2006 and now serving over 9.5 million customers across 180+ countries. Its production is certified to ISO 9001 (quality management) and ISO 14001 (environmental management), with RoHS-compliant materials and processes throughout. Every board is reviewed before production and tested (automated optical inspection, electrical probe testing, and a final manual check) before shipping. This scale and standardization are what make a low-volume, custom board like BOLOTIE's financially viable at the target unit cost.

France: do the same for the Europe-based manufacturer the camera-mounting case, produced locally by [local manufacturer name] to limit the transport footprint of the product's bulkiest component.

PCB, CNC, plastic covers, motor

C4 : Présenter les principaux impacts environnementaux du projet présenté :

- identification des postes d'impact majeurs du projet : déplacements et mobilité des équipes et des publics, consommation énergétique, équipements mobilisés, données et flux numériques, usages et exploitation.
- méthode d'identification des impacts (analyse qualitative, ordres de grandeur, indicateurs)

As stated in B6, BOLOTIE has two main positive environmental impacts. First, it is designed with high-quality, long-lasting materials to ensure the longevity of the units, coupled with the open-source publication of the software — together, these are expected to keep each unit functional for decades. Second, BOLOTIE may restore functionality to old equipment currently out of use due to spring-motor failure, avoiding both the manufacturing footprint of a replacement and the disposal footprint of the broken camera.

The main negative impact identified concerns production and shipping, arising from the choice of a Chinese manufacturer for the electronic core. This is offset, as detailed in B6, by batch production, optimized freight logistics, and the use of durable, high-tier components to limit the need for repeat shipments. Throughout the 18 months impacts are assessed qualitatively, optimizing shipping quantities and the durability of the pieces chosen. Finally, during the normal commercialization phase, and depending on the product's sales performance, these choices will be reassessed to examine whether relocating production to Europe becomes financially feasible.

A second, minor impact concerns travel for promotional purposes. To minimize this, train travel will be prioritized for any distance under 12 hours. For longer trips, such as transatlantic travel to the United States, participation in multiple events will be planned around a single trip wherever possible, to reduce both the number of flights and associated costs. The website, social media presence, and GitHub-hosted firmware all have a (typically small but non-zero) hosting/data footprint.

Finally, BOLOTIE comes at a moment when renewed interest in film also calls for smarter technological solutions that extend the life of what already works. By keeping decades-old cameras in active use, and augmenting their useability toward a wider range of productions,

BOLOTIE offers a concrete, replicable, and community-driven example of how technical innovation can serve sustainability in analog filmmaking, while expanding the kind of cinema that can be made on film today.

Annexes financières et administratives

Vous pouvez porter ici des justifications/éclaircissements portant sur les annexes financières, dont le modèle est joint.

Merci de ne pas envoyer les éléments en un seul fichier PDF. Des sous dossiers (administratif/projet/...) sont possibles et appréciés.

Autres soutiens publics reçus par l'entreprise sur les 3 dernières années (y compris autres projets)

- Autres soutiens publics reçus par l'entreprise et objets précis de ces soutiens, autres demandes en cours ou antérieures au titre de ce projet
- Soutien de l'IFCIC, soutien de Bpifrance, soutien du CNC (FAJV, Création immersive, Aides financières à la V&D, Choc de modernisation, Grande fabrique de l'image...), soutien du FUI, du FCE, crédit d'impôt (CIR, CIIJV), soutien de l'ANR...
- Soutiens européens
- JEI ou autre exonération de charges
- Labellisation du projet par des pôles de compétitivité

Éléments obligatoires à transmettre

- **L'ensemble des devis**
- **Le fichier Descriptif financier dûment rempli,**
- **Un organigramme capitalistique de l'entreprise avec l'ensemble des actionnaires moraux et physiques et leurs éventuelles parts détenues dans d'autres sociétés, leurs chiffres d'affaires, effectifs et activités respectives,**
- **Un extrait K-BIS de moins de trois mois,**
- Si l'entreprise a en présidence une/des entreprise(s), joindre un Kbis de moins de 3 mois correspondant,
- **Les trois dernières liasses fiscales,**
- **Les derniers statuts de l'entreprise,**
- **Les attestations fiscales et sociales (URSSAF, AUDIENS, POLE EMPLOI) datant de moins de six mois,**
- **L'attestation de formation « Prévenir et agir contre les violences sexistes et sexuelles » dispensée par le CNC**
- Un RIB dématérialisé à l'un des formats suivants :
 - Un scan tamponné et signé par la banque, au format PDF non modifiable
 - Ou un scan ou un document PDF téléchargé en ligne, accompagné d'une attestation du représentant légal : l'attestation peut être transmise au gestionnaire grâce à un mail, qui permet de bien identifier son émetteur en tant que représentant légal, ou via un document PDF, et doit respecter la formule type suivante :

« Je soussigné M ou Mme NOM Prénom représentant légal⁸ de la société XXX atteste que le compte bancaire en pièce jointe est ouvert au nom de la société XXX SIREN 999 999 999 »

Monsieur/Madame NOM Prénom Ex : Gérant/Président de la société

Si le signataire n'est pas le représentant statutaire ou légal de l'association/la société/la commune, joindre le pouvoir ou mandat (portant les deux signatures : celles du représentant légal et celle de la personne qui va le représenter) lui permettant d'engager celle-ci.